

## **Rising Optimism and Lingering Anxiety:**

The Public Perception of AI Through Bluesky Posts in Work and Education

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### **Introduction & Motivation: Why this research question matters for social science**

The landscape of digital interaction and information consumption has been reshaped dramatically following the public debut of ChatGPT on November 30, 2022. In a historically short period, Generative AI has moved from a novelty to a ubiquitous utility, with ChatGPT currently ranking among the top 5 most visited websites globally. This is not a monopoly, but a shift; major companies including Gemini, Claude, and Deepseek are aggressively competing to permeate both the technology sector and our daily lives.

While the adoption metrics are clear, the societal reaction is more complex. Social science is fundamentally concerned with how human populations adapt to external shocks, and the rapid rise of AI represents a significant disruption. The central motivation of this research is to look past the usage statistics to understand the "real feelings" of the general public. We aim to determine the dominant emotional response to this technological integration: is it driven by excitement for new capabilities, or fear of the unknown?

This question is critical for social science because AI is not just affecting leisure; it is impacting the foundational pillars of society: Work and Education.

- **Labor Market Anxiety:** A key area of sociological interest is the public narrative regarding labor substitution. Our research investigates the specific anxiety that AI will replace human workers. Understanding whether this is a dominant public sentiment or a fringe concern is vital for policymakers and economists.
- **Perception of Risk vs. Optimism:** By quantifying "Net Risk" (measuring fear and nervousness against optimism), we can track the psychological toll of technological acceleration.
- **Temporal Adaptation:** Finally, this research matters because public sentiment is not static. We are observing a temporal evolution where, over time, the public may be viewing AI more positively as familiarity grows. Understanding this trajectory—how humans normalize powerful new tools—is a core contribution to the study of social adaptation.

To capture this organic public sentiment, this study moves beyond traditional surveys, utilizing data collection via the Bluesky API to analyze the discourse in realtime.

### **Data: Source, sampling, cleaning steps, descriptive statistics, limitations.**

We collected our data using the Bluesky API. We wrote a python script that called on the Bluesky API to collect all posts that contained the keywords like ChatGPT, AI, and LLM. In collecting the data we got the keyword used to find the post, author, post text, when the post was created, likes, reposts, replies, authors name, and whether it has embedded media. We cleaned the data in a couple of different ways for different things that we were using the data for. For the sentiment analysis we cleaned our data by getting rid of urls peoples account information (someone @ a friend in a post). We also made the emojis words and made every letter lowercase. When doing the BERTopic it was more trial and error. We tried many things and got rid of stopwords as well as words like and, to, of, because they just ended up being the topics rather than what we were actually trying to get at.

These are the descriptive statistics for our dataset. As you can see we collected an extremely high number of posts. This relates to one of our main limitations which was just the size of the dataset. The dataset is so large that it is very difficult to manage and it made it hard to run things because there were just so many posts. Another limitation of this dataset is where we collected it. We have a limited view because we are only capturing the

```
===== DESCRIPTIVE STATISTICS =====
Total posts collected: 983828
Date range: 2022-11-30 00:29:50+00:00 to 2025-10-30 01:39:08.378000+00:00
Unique authors: 255742
Percent with embedded media: 49.33%
Posts removed during cleaning: raw_len not provided.

--- Word Count Statistics ---
Median words per post: 23.0
Mean words per post: 24.68

Full distribution:
count      983828.000000
mean       24.684810
std        16.000218
min         0.000000
25%        11.000000
50%        23.000000
75%        38.000000
max       103.000000
Name: word_count, dtype: float64
```

opinions of people who are using Bluesky. This is a limitation because not everyone has Bluesky and it attracts specific people who use the app. Beyond just the people who use Bluesky there is some bias in who is going to be talking about AI. These may be people who feel more strongly about AI for any positive or negative reason. While this is capturing peoples naturally occurring opinions about AI there is still bias in where the data is collected and who is represented in our dataset.

### **Methods: Exact preprocessing steps, models used, parameter choices, evaluation metrics.**

We collected our dataset using a custom Python scraper built with the official Bluesky API through the atproto library. The script logged in using stored credentials and repeatedly searched

for AI related keywords, these words were “chatgpt,” “openai,” “llm,” “ai,” “gemini,” and “deepseek”. These were chosen to capture a wide range of posts about generative AI. Because the Bluesky API limits how far back posts can be retrieved, we divided the timeline into 30 day windows starting on November 30, 2022. For each window, the scraper requested up to 100 posts at a time and used cursor based pagination until no additional posts were available, ensuring full coverage across the entire time period. While scraping, we ran into an issue with an outdated API field related to image metadata. To avoid losing data, we added a small wrapper that caught validation errors, downloaded the raw JSON, fixed the incorrect field, and then continued processing. We also built in automatic rate limit handling: the scraper tracked how many calls were made within each five minute interval, paused when approaching Bluesky’s limit of 60 calls, and saved progress before waiting. If the server temporarily failed, the script wrote a checkpoint, paused for five minutes, and then resumed. These steps made the scraper stable during long runs.

For each post, we kept the text, timestamp, author ID, engagement metrics, the keyword that surfaced the post, and whether the post contained media. Since some posts appeared under multiple keywords, we removed duplicates using the post’s URI as a unique identifier. All posts were saved in a single JSON file, which was periodically overwritten through checkpointing to protect against data loss.

After data collection, we implemented separate preprocessing pipelines for sentiment analysis and topic modeling. For sentiment analysis, we performed minimal text cleaning to preserve emotional nuance: all text was lowercased, URLs and user handles were removed, and emojis were converted into textual descriptors. We then applied VADER (Valence Aware Dictionary and Sentiment Reasoner), a sentiment classifier widely used for short, informal text such as social media posts. VADER produces four scores—positive, negative, neutral, and a compound score representing the overall sentiment intensity. We used the compound score as our primary sentiment metric because it captures both direction and intensity on a standardized scale. This method was chosen because VADER performs reliably on short form content and does not require extensive computational resources.

For topic modeling, we used BERTopic, which integrates contextual embeddings, dimensionality reduction, and density-based clustering to identify coherent themes within large text corpora. BERTopic requires more aggressive cleaning than VADER, so we removed stopwords,

constructed a small custom stoplist to eliminate extremely common but uninformative terms (including “ai,” “chatgpt,” and “llm”), and lemmatized text using SpaCy to reduce vocabulary sparsity. Very short posts were removed because they tended to hinder cluster formation. BERTopic generated embeddings using the default sentence transformer model, reduced them using UMAP, and clustered them using HDBSCAN, producing a set of interpretable topics with corresponding representative keywords and exemplar posts.

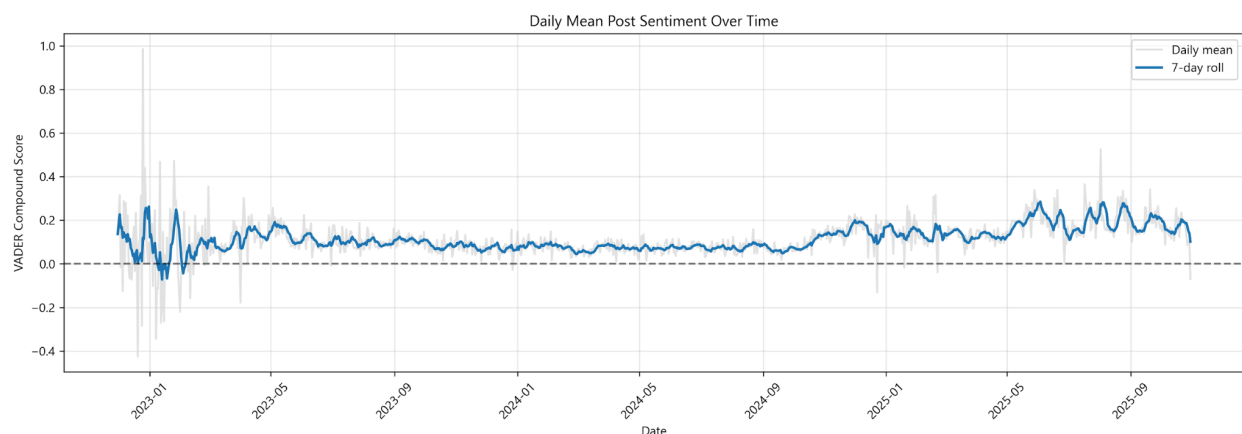
Together, these methods allowed us to capture both the emotions of AI related public discourse and the underlying thematic structure driving discussion trends on Bluesky.

## **Results: Public Perception of AI in Work and Education**

The analysis of public sentiment regarding artificial intelligence, specifically within the domains of work and education, revealed several key quantitative trends. The results are derived from topic modeling and sentiment analysis performed on data collected via the Bluesky API.

### **1. Temporal Evolution of General Sentiment**

The longitudinal analysis of daily mean post sentiment indicates a shift in public perception over the studied period.



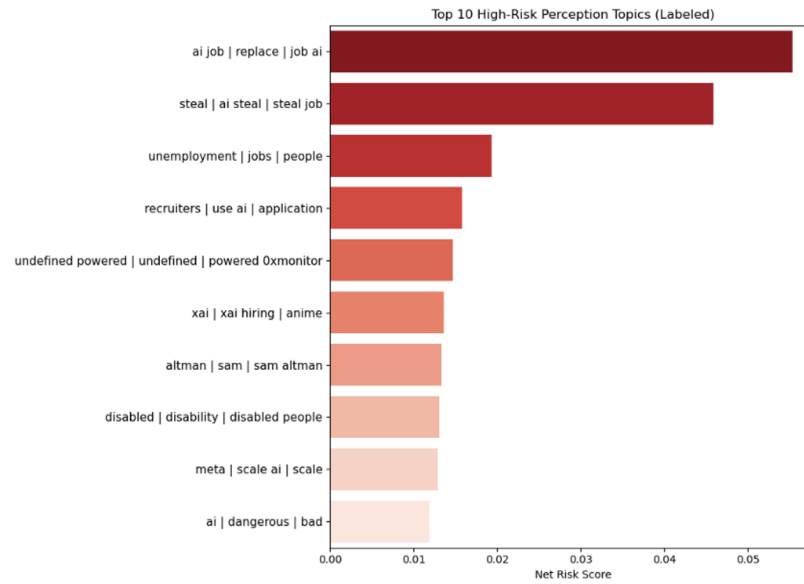
- **Sentiment Trends:** Early data from late 2022 and early 2023 shows high volatility in sentiment scores. However, moving into 2024 and 2025, the "Daily Mean Post Sentiment" stabilizes and exhibits a gradual upward trend, suggesting a normalization of AI discourse.

### **2. Risk Perception and Fear Analysis**

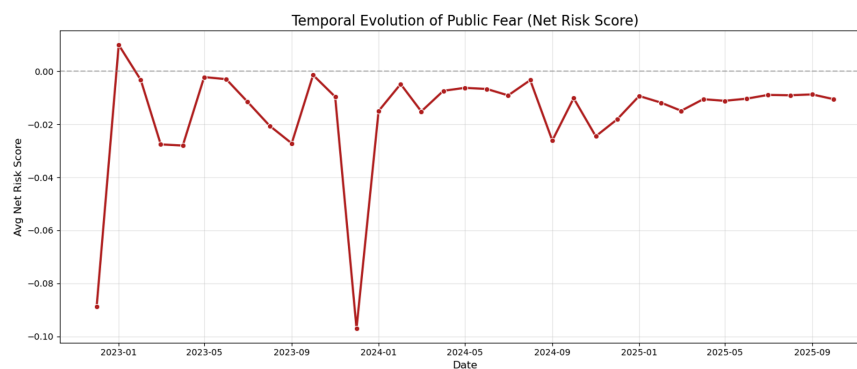
To quantify public anxiety, a "Net Risk Score" was calculated using the following formula:

$$\text{Net Risk} = (\text{Fear} + \text{Nervousness}) - \text{Optimism}$$

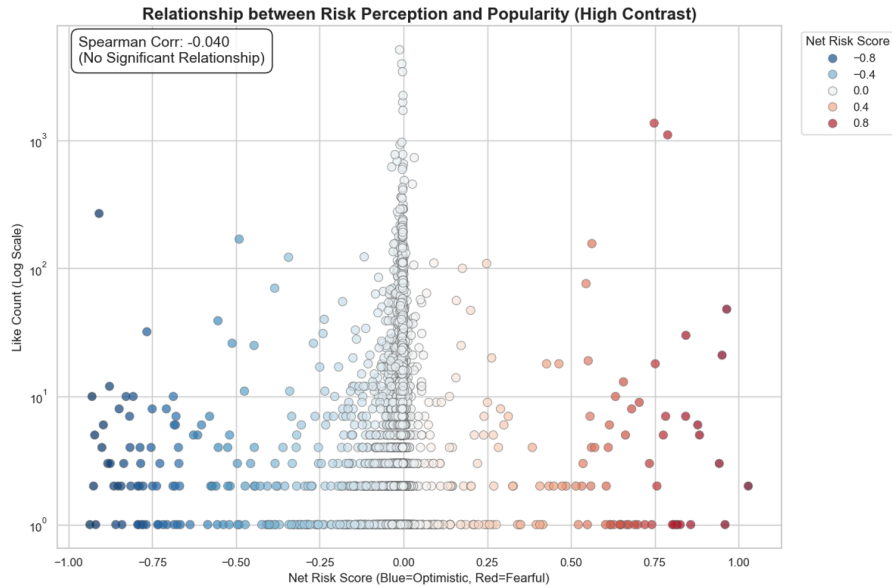
- **Top High-Risk Topics:** Topic modeling identified specific themes driving high-risk perceptions. The topic labeled "ai job | replace | job ai" yielded the highest risk score, indicating that the fear of workforce displacement is the primary driver of negative sentiment. Other high-risk topics included concerns about theft ("steal | ai steal") and general unemployment.



- **Temporal Risk Evolution:** The "Temporal Evolution of Public Fear" (Avg Net Risk Score) graph demonstrates that while there were spikes in fear (positive net risk) in early 2023, the score largely fluctuated below zero for the remainder of the timeline. This implies that, on average, optimism often outweighed fear and nervousness.

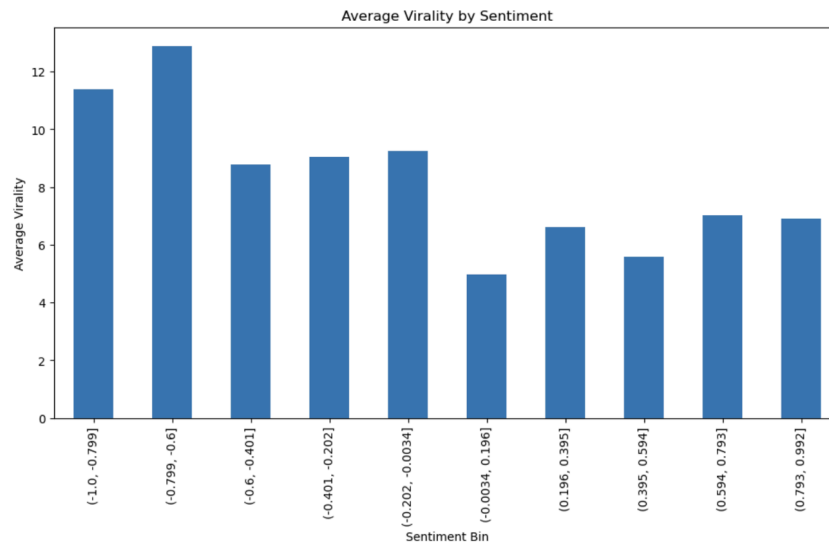


- **Correlation with Popularity:** A Spearman correlation analysis was conducted to test if fear drives engagement. The result yielded a Spearman Correlation of -0.040, indicating no significant relationship between the Net Risk Score of a post and its "Like Count".



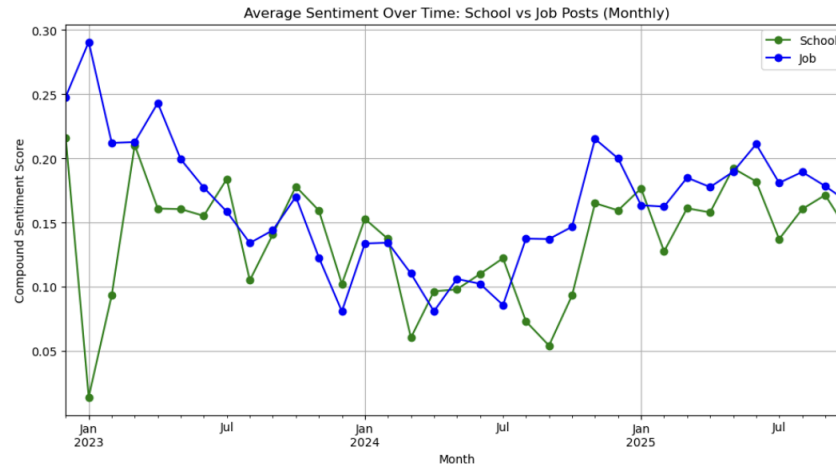
### 3. Virality and Sentiment Intensity

While general popularity did not correlate with risk scores, specific sentiment intensity bins showed distinct virality patterns.



- **Negative Virality:** Posts falling into the most extreme negative sentiment bin (-1.0 to -0.799) achieved the highest "Average Virality" score (approximately 11.5), followed closely by the second most negative bin.
- **Engagement Gap:** Neutral and moderately positive posts showed significantly lower average virality compared to extreme negative content, suggesting that negative content generates more intense individual engagement.

#### 4. Domain Comparison: Work vs. Education



Comparative analysis between job-related and school-related datasets highlighted distinct sentiment trajectories.

- **Sentiment Co-movement:** The "Average Sentiment Over Time: School vs Job Posts" figure shows that sentiment in both domains often tracks together. However, school-related sentiment demonstrated higher volatility, particularly in early 2023 where it dropped significantly while job sentiment experienced a significant spike.
- **Education Optimism:** Similar to the general trend, the specific filter for school-related words showed a massive increase in positive label counts starting in mid-2024, far outpacing negative or neutral educational discussions.

#### Error Analysis & Statistical Correlations

We tested the relationship between "Net Risk Score" and "Like Count" (popularity) to see if fear-mongering drove engagement. The Spearman Correlation was -0.040, indicating no significant statistical relationship between the level of risk expressed in a post and its popularity. This suggests that while negative sentiment is viral, specific risk narratives do not linearly correlate with likes.

#### Interpretation: What the results mean in context

The data suggests that the initial "shock" of generative AI is wearing off. The upward trend in sentiment and the massive increase in "neutral" post volume indicate that AI is transitioning from a novelty-inducing panic to a mundane utility. As technology permeates daily life, the public is becoming desensitized, moving from emotional reaction to practical discussion.

While general sentiment is improving, specific anxieties remain acute. The interpretation of the topic modeling confirms that public nervousness is not abstract; it is economically grounded. The prominence of keywords like "replace," "workers," and "unemployment" demonstrates that people are nervous, it is mostly focused on AI replacing them, specifically on economical survival.

The finding that negative sentiment drives the highest virality aligns with the "negativity bias" often seen in social media algorithms. However, the lack of correlation between "Net Risk" and popularity implies a nuance: users engage with general negativity/complaints, but specific, high-risk fear-mongering does not necessarily guarantee high engagement.

Data indicates a persistent "sentiment gap" favoring the labor market over education. While both categories saw a recovery in 2025, job-related sentiment consistently outperforms school-related sentiment. The educational sentiment experienced deeper troughs, particularly in the middle 2024, suggesting that the academic environment is currently viewed with more frustration or stress than the professional sphere. Rather than the classroom being a refuge, the workplace appears to be the source of greater relative optimism.

### **Social Science Implications**

These findings challenge the narrative of inherent "techno-phobia." The longitudinal data implies that human adaptability is rapid. The shift from fear to neutrality/positivity suggests that societies "metabolize" disruptive technologies quickly. As users engage with the tools, the abstract fear of the unknown is replaced by a concrete understanding of utility.

The persistence of labor-replacement topics as the highest risk category implies that AI is fundamentally reshaping the cultural value of work. The discourse is not just about tools, but about value and identity. Socially, this indicates that institutions (corporations, governments) have failed to assuage fears regarding the human place in the future economy, even as the technology itself becomes popular.

The virality data has implications for how public opinion is formed. If the most visible posts (highest virality) are the most negative, but the majority of posts are neutral or positive, there is a distortion in public perception. Social media platforms may be amplifying a "fear signal" that does not accurately reflect the aggregate "true feeling" of the quiet majority. This disconnect between visible discourse (viral fear) and actual sentiment (rising positivity) is a critical area for understanding digital sociology.